

**Online Resource 7 -Trace element concentrations in sulphide minerals**

**Table 1: Trace element concentrations in sphalerite. For calibration MUL-ZnS-1 was used for most elements. For elements marked with \*USGS MASS-1, with \*\*NIST SRM 610 was applied as calibration material. All values are given in µg/g. Also given are obtained trace element concentrations for analysed quality control materials, together with respective reference values if available.**

| Sample  | Spot | Mn   | Fe   | Co   | Ni   | Cu   | Ga   | Ge   | As   | Se   | Ag   | Cd  | In   | Sn   | Sb   | Au** | Hg*  | Tl   | Pb   | Bi*  |
|---------|------|------|------|------|------|------|------|------|------|------|------|-----|------|------|------|------|------|------|------|------|
| 16B07   | 11   | 7.21 | 1330 | bdl  | 0.28 | 1470 | 14.6 | 1.01 | 40.8 | bdl  | 41.5 | 965 | 3.33 | 650  | 32.6 | 0.10 | 2650 | 0.44 | 99.3 | 0.04 |
| 16B07   | 12   | 5.58 | 1440 | bdl  | bdl  | 733  | 17.6 | 0.59 | 48.1 | bdl  | 38.2 | 947 | 2.28 | 495  | 113  | 0.13 | 2550 | 0.52 | 93.6 | 0.02 |
| 16B07   | 16   | 6.37 | 1120 | bdl  | bdl  | 537  | 51.5 | 3.18 | 36.1 | bdl  | 27.5 | 969 | 0.98 | 309  | 44.9 | 0.19 | 2890 | 0.23 | 98.8 | 0.03 |
| 16B07   | 17   | 8.67 | 1390 | bdl  | bdl  | 1500 | 31.5 | 1.94 | 57.6 | bdl  | 33.2 | 906 | 4.00 | 1160 | 52.3 | 0.06 | 2770 | 0.51 | 60.0 | 0.07 |
| 16B07   | 18   | 7.07 | 1300 | bdl  | bdl  | 693  | 49.8 | 2.84 | 42.3 | bdl  | 26.9 | 970 | 2.72 | 771  | 41.7 | 0.18 | 2990 | 0.17 | 94.8 | 0.03 |
| 16B07   | 19   | 12.4 | 1250 | bdl  | bdl  | 1320 | 15.4 | 1.58 | 50.1 | bdl  | 93.0 | 876 | 2.15 | 583  | 67.2 | 0.09 | 2440 | 0.47 | 91.6 | 0.03 |
| 16B07   | 22   | 9.47 | 1430 | bdl  | bdl  | 1470 | 176  | 15.9 | 44.8 | bdl  | 57.2 | 928 | 3.99 | 1110 | 87.4 | 0.09 | 2770 | 0.46 | 55.3 | 0.03 |
| 16B07   | 23   | 9.20 | 1460 | bdl  | 0.12 | 1940 | 174  | 19.1 | 45.4 | bdl  | 74.9 | 973 | 4.51 | 1230 | 89.0 | 0.13 | 3320 | 0.34 | 74.5 | 0.04 |
| 16B07   | 26   | 7.82 | 1060 | 0.13 | 0.21 | 626  | 51.7 | 2.85 | 27.7 | bdl  | 41.8 | 806 | 1.13 | 255  | 37.1 | 0.19 | 2590 | 0.39 | 71.6 | 0.02 |
| 16B07   | 27   | 9.57 | 1200 | 0.11 | 0.09 | 452  | 14.9 | 0.93 | 13.1 | bdl  | 34.8 | 902 | 1.11 | 246  | 29.8 | 0.15 | 2690 | 0.18 | 91.1 | 0.03 |
| 16B07   | 29   | 10.6 | 1230 | 0.08 | bdl  | 244  | 13.8 | 0.78 | 30.8 | 22.4 | 17.2 | 940 | 0.72 | 141  | 23.2 | 0.06 | 2880 | 0.18 | 47.3 | 0.03 |
| 16B07   | 31   | 6.33 | 1050 | bdl  | bdl  | 283  | 4.98 | bdl  | 8.10 | 38.5 | 25.7 | 920 | 0.25 | 1.47 | 18.7 | 0.08 | 2610 | 0.13 | 43.9 | 0.05 |
| 16B07   | 32   | 6.50 | 1090 | bdl  | bdl  | 262  | 5.89 | 0.79 | 26.7 | 11.8 | 22.3 | 898 | 0.19 | 20.1 | 19.2 | 0.15 | 2600 | 0.14 | 54.4 | 0.02 |
| 16B07   | 33   | 5.25 | 1020 | bdl  | bdl  | 282  | 5.01 | bdl  | 25.5 | 47.7 | 32.8 | 859 | 0.25 | 24.6 | 62.8 | 0.09 | 2760 | 0.38 | 79.5 | 0.02 |
| 16B07   | 35   | 6.56 | 1200 | bdl  | bdl  | 402  | 11.0 | 1.44 | 37.2 | 18.6 | 34.6 | 930 | 0.46 | 69.9 | 70.9 | 0.05 | 3060 | 0.40 | 112  | 0.03 |
| 16B07   | 38   | 22.8 | 1190 | bdl  | bdl  | 587  | 10.1 | 0.29 | 42.0 | bdl  | 56.2 | 805 | 0.52 | 44.2 | 56.1 | 0.18 | 2220 | 0.49 | 173  | 0.02 |
| 16B07   | 39   | 12.2 | 1130 | bdl  | bdl  | 390  | 10.9 | 0.35 | 34.9 | bdl  | 33.5 | 965 | 0.53 | 50.6 | 34.9 | 0.08 | 2860 | 0.20 | 63.1 | 0.03 |
| 16B06   | 58   | 9.54 | 1170 | bdl  | bdl  | 238  | 4.21 | 0.41 | bdl  | bdl  | 14.8 | 821 | 0.47 | 1.67 | 7.02 | 0.05 | 2240 | 0.09 | 41.4 | 0.03 |
| 16B06   | 59   | 10.5 | 1270 | bdl  | bdl  | 368  | 6.24 | 0.29 | bdl  | bdl  | 14.7 | 838 | 0.40 | 1.13 | 7.00 | 0.10 | 2360 | 0.07 | 46.4 | 0.02 |
| 16B06   | 62   | 18.0 | 1220 | bdl  | bdl  | 250  | 5.47 | 0.22 | bdl  | 12.1 | 17.4 | 841 | 0.82 | 1.39 | 6.34 | 0.16 | 2360 | 0.29 | 421  | 0.02 |
| 16B06   | 63   | 13.9 | 1310 | bdl  | bdl  | 137  | 6.05 | 0.41 | bdl  | bdl  | 7.51 | 868 | 0.73 | 41.6 | 4.35 | 0.06 | 3050 | 0.21 | 26.0 | 0.04 |
| 16B06   | 65   | 16.5 | 1170 | bdl  | bdl  | 291  | 6.62 | 0.48 | 16.2 | bdl  | 16.8 | 831 | 0.58 | 50.0 | 16.4 | 0.14 | 2600 | 0.13 | 76.1 | 0.02 |
| 16B06   | 68   | 8.97 | 1050 | bdl  | bdl  | 191  | 8.52 | 0.81 | 9.59 | bdl  | 18.1 | 823 | 0.54 | 27.2 | 12.2 | 0.14 | 2500 | 0.25 | 69.2 | 0.02 |
| 16B06   | 69   | 16.7 | 1070 | bdl  | 0.26 | 320  | 7.19 | bdl  | bdl  | bdl  | 21.7 | 811 | 0.61 | 1.67 | 15.1 | 0.06 | 2350 | 0.38 | 115  | 0.03 |
| 16B06   | 70   | 38.8 | 1060 | bdl  | 0.09 | 224  | 5.69 | bdl  | 19.0 | bdl  | 18.5 | 840 | 0.33 | 3.13 | 23.0 | 0.39 | 2500 | 0.81 | 207  | 0.06 |
| 16B06   | 72   | 11.2 | 1200 | 0.4  | 0.10 | 123  | 1.87 | bdl  | 6.70 | 17.2 | 9.6  | 850 | 0.62 | 0.48 | 9.21 | 0.11 | 2880 | 0.28 | 42.0 | 0.02 |
| 16B09_2 | 73   | 11.5 | 1370 | bdl  | bdl  | 144  | 6.53 | 0.31 | 25.9 | 12.9 | 15.7 | 983 | 0.73 | 15.3 | 15.7 | 0.06 | 5210 | 0.80 | 45.1 | 0.04 |
| 16B09_2 | 91   | 14.1 | 1290 | bdl  | bdl  | 207  | 8.64 | bdl  | 9.38 | bdl  | 15.5 | 963 | 0.33 | 1.69 | 24.7 | 0.08 | 3830 | 0.45 | 100  | 0.02 |
| 16B09_2 | 92   | 22.6 | 1350 | bdl  | bdl  | 770  | 9.88 | 4.08 | 36.9 | 17.0 | 41.3 | 972 | 2.17 | 452  | 186  | 0.04 | 3940 | 1.22 | 14.7 | 0.10 |

| 16B09_2 | 96   | 16.3 | 1290 | bdl  | bdl  | 534  | 4.52 | 7.34 | 25.3 | bdl  | 29.2 | 994  | 1.64 | 227  | 80.7 | 0.10 | 3680 | 0.70 | 334  | 0.05 |
|---------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 16B09_2 | 98   | 10.9 | 1490 | bdl  | bdl  | 1510 | 4.42 | 0.57 | 43.7 | bdl  | 23.9 | 979  | 6.38 | 1540 | 87.5 | 0.10 | 4460 | 0.67 | 75.5 | 0.04 |
| 16B09_2 | 99   | 11.8 | 1450 | bdl  | bdl  | 942  | 4.80 | bdl  | 59.4 | 11.9 | 20.9 | 945  | 3.32 | 744  | 40.2 | 0.05 | 4390 | 0.96 | 66.0 | 0.03 |
| 16B09_2 | 102  | 20.9 | 1580 | bdl  | bdl  | 1790 | 33.9 | 6.20 | 42.0 | bdl  | 61.6 | 882  | 9.14 | 2470 | 364  | 0.10 | 4240 | 1.51 | 443  | 0.03 |
| 16B09_2 | 103  | 16.4 | 1560 | bdl  | bdl  | 369  | 13.9 | 1.02 | 33.3 | bdl  | 22.8 | 949  | 0.67 | 51.8 | 35.9 | 0.10 | 4500 | 0.94 | 179  | 0.03 |
| 16B09_2 | 105  | 15.5 | 1580 | bdl  | bdl  | 806  | 42.6 | 2.35 | 57.0 | bdl  | 27.9 | 958  | 1.90 | 503  | 132  | 0.03 | 5420 | 1.75 | 119  | 0.03 |
| 16B09_2 | 107  | 12.3 | 1490 | bdl  | bdl  | 524  | 55.8 | 2.30 | 67.0 | bdl  | 35.0 | 1010 | 2.08 | 122  | 206  | bdl  | 5060 | 1.40 | 65.1 | 0.04 |
| 16B09_2 | 109  | 10.1 | 1340 | bdl  | bdl  | 371  | 7.01 | 0.38 | 47.4 | bdl  | 22.2 | 993  | 2.43 | 93.3 | 53.6 | 0.02 | 4380 | 0.99 | 91.0 | 0.03 |
|         |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 1 SD    |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
| Sample  | Spot | Mn   | Fe   | Co   | Ni   | Cu   | Ga   | Ge   | As   | Se   | Ag   | Cd   | In   | Sn   | Sb   | Au** | Hg*  | Tl   | Pb   | Bi*  |
| 16B07   | 11   | 0.51 | 110  | 0.03 | 0.10 | 60.1 | 2.28 | 0.20 | 4.62 | 4.32 | 1.53 | 35.5 | 0.16 | 43.5 | 2.12 | 0.04 | 572  | 0.07 | 4.84 | 0.02 |
| 16B07   | 12   | 0.42 | 120  | 0.04 | 0.06 | 30.2 | 2.76 | 0.18 | 5.40 | 4.57 | 1.47 | 35.2 | 0.13 | 33.2 | 7.15 | 0.05 | 551  | 0.08 | 4.59 | 0.02 |
| 16B07   | 16   | 0.50 | 94.7 | 0.05 | 0.03 | 22.6 | 8.02 | 0.52 | 4.89 | 5.01 | 1.18 | 36.3 | 0.09 | 20.8 | 3.01 | 0.07 | 624  | 0.07 | 4.88 | 0.02 |
| 16B07   | 17   | 0.67 | 117  | 0.05 | 0.08 | 61.9 | 4.95 | 0.39 | 6.80 | 5.67 | 1.43 | 34.3 | 0.24 | 77.8 | 3.53 | 0.04 | 597  | 0.11 | 3.04 | 0.03 |
| 16B07   | 18   | 0.53 | 109  | 0.04 | 0.10 | 28.8 | 7.74 | 0.47 | 5.38 | 5.23 | 1.12 | 36.2 | 0.16 | 51.7 | 2.78 | 0.06 | 645  | 0.05 | 4.67 | 0.02 |
| 16B07   | 19   | 0.83 | 103  | 0.03 | 0.06 | 53.6 | 2.40 | 0.26 | 5.37 | 3.93 | 3.26 | 32.2 | 0.11 | 39.0 | 4.24 | 0.03 | 526  | 0.06 | 4.45 | 0.01 |
| 16B07   | 22   | 0.71 | 120  | 0.04 | 0.08 | 60.7 | 27.3 | 2.16 | 5.70 | 5.11 | 2.23 | 35.0 | 0.23 | 74.1 | 5.68 | 0.04 | 598  | 0.10 | 2.80 | 0.01 |
| 16B07   | 23   | 0.66 | 121  | 0.05 | 0.07 | 79.4 | 26.8 | 2.53 | 5.89 | 5.78 | 2.74 | 36.1 | 0.23 | 82.1 | 5.68 | 0.05 | 716  | 0.07 | 3.67 | 0.02 |
| 16B07   | 26   | 0.60 | 90.2 | 0.06 | 0.14 | 26.3 | 8.05 | 0.49 | 4.37 | 4.98 | 1.71 | 30.6 | 0.10 | 17.3 | 2.56 | 0.07 | 559  | 0.09 | 3.59 | 0.02 |
| 16B07   | 27   | 0.66 | 99.6 | 0.04 | 0.06 | 18.7 | 2.34 | 0.19 | 3.14 | 4.59 | 1.31 | 33.3 | 0.07 | 16.5 | 1.96 | 0.05 | 581  | 0.04 | 4.44 | 0.01 |
| 16B07   | 29   | 0.78 | 104  | 0.05 | 0.03 | 10.8 | 2.22 | 0.22 | 4.96 | 6.58 | 0.84 | 35.4 | 0.08 | 9.65 | 1.70 | 0.03 | 621  | 0.06 | 2.42 | 0.02 |
| 16B07   | 31   | 0.48 | 88.2 | 0.04 | 0.06 | 12.1 | 0.83 | 0.15 | 3.40 | 5.95 | 1.06 | 34.3 | 0.04 | 0.20 | 1.33 | 0.03 | 563  | 0.05 | 2.21 | 0.02 |
| 16B07   | 32   | 0.57 | 94.9 | 0.05 | bdl  | 12.1 | 1.05 | 0.28 | 5.21 | 6.85 | 1.16 | 34.8 | 0.05 | 1.66 | 1.58 | 0.07 | 562  | 0.07 | 2.85 | 0.01 |
| 16B07   | 33   | 0.40 | 85.0 | 0.04 | 0.02 | 11.9 | 0.82 | 0.12 | 3.77 | 5.67 | 1.26 | 31.8 | 0.03 | 1.74 | 4.02 | 0.03 | 595  | 0.07 | 3.89 | 0.01 |
| 16B07   | 35   | 0.53 | 102  | 0.05 | 0.08 | 17.3 | 1.79 | 0.33 | 5.57 | 6.73 | 1.47 | 35.2 | 0.06 | 4.89 | 4.68 | 0.03 | 661  | 0.09 | 5.56 | 0.02 |
| 16B07   | 38   | 1.55 | 99.6 | 0.03 | 0.07 | 24.5 | 1.63 | 0.14 | 5.30 | 4.73 | 2.14 | 30.3 | 0.06 | 3.12 | 3.68 | 0.07 | 480  | 0.09 | 8.45 | 0.02 |
| 16B07   | 39   | 0.81 | 93.2 | 0.03 | 0.03 | 16.0 | 1.70 | 0.12 | 4.21 | 4.10 | 1.22 | 35.3 | 0.04 | 3.43 | 2.23 | 0.03 | 618  | 0.04 | 3.07 | 0.01 |
| 16B06   | 58   | 0.72 | 99.7 | 0.05 | 0.04 | 10.6 | 0.74 | 0.17 | 3.89 | 6.34 | 0.77 | 31.2 | 0.06 | 0.26 | 0.69 | 0.03 | 483  | 0.05 | 2.14 | 0.02 |
| 16B06   | 59   | 0.78 | 108  | 0.05 | 0.09 | 15.9 | 1.06 | 0.14 | 3.53 | 5.47 | 0.77 | 31.9 | 0.06 | 0.21 | 0.69 | 0.05 | 510  | 0.04 | 2.39 | 0.01 |
| 16B06   | 62   | 1.29 | 105  | 0.04 | 0.21 | 11.3 | 0.96 | 0.16 | 3.64 | 6.61 | 0.92 | 32.3 | 0.10 | 0.25 | 0.69 | 0.07 | 510  | 0.09 | 20.5 | 0.02 |
| 16B06   | 63   | 0.97 | 109  | 0.05 | 0.10 | 6.18 | 0.99 | 0.16 | 3.69 | 6.23 | 0.41 | 32.4 | 0.07 | 2.92 | 0.46 | 0.03 | 657  | 0.06 | 1.35 | 0.02 |
| 16B06   | 65   | 1.25 | 102  | 0.04 | 0.14 | 13.6 | 1.19 | 0.22 | 5.01 | 7.36 | 1.01 | 32.7 | 0.09 | 3.73 | 1.47 | 0.08 | 561  | 0.07 | 3.94 | 0.03 |

|                |     |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |
|----------------|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| <b>16B06</b>   | 68  | 0.67 | 89.1 | 0.04 | 0.05 | 8.58 | 1.39 | 0.21 | 3.59 | 5.70 | 0.85 | 31.1 | 0.06 | 2.01 | 0.99 | 0.06 | 539  | 0.07 | 3.46 | 0.02 |
| <b>16B06</b>   | 69  | 1.15 | 90.0 | 0.04 | 0.14 | 13.6 | 1.17 | 0.14 | 3.80 | 6.17 | 0.93 | 30.4 | 0.06 | 0.23 | 1.13 | 0.03 | 507  | 0.08 | 5.66 | 0.02 |
| <b>16B06</b>   | 70  | 2.68 | 93.0 | 0.05 | 0.12 | 10.7 | 1.03 | 0.10 | 5.19 | 7.53 | 1.04 | 32.8 | 0.07 | 0.46 | 1.86 | 0.14 | 541  | 0.17 | 10.2 | 0.04 |
| <b>16B06</b>   | 72  | 0.80 | 101  | 0.09 | 0.07 | 5.70 | 0.35 | 0.10 | 3.36 | 5.76 | 0.51 | 31.9 | 0.06 | 0.12 | 0.77 | 0.04 | 622  | 0.07 | 2.13 | 0.02 |
| <b>16B09_2</b> | 73  | 0.78 | 112  | 0.04 | 0.08 | 6.12 | 1.04 | 0.11 | 3.96 | 5.03 | 0.62 | 36.1 | 0.05 | 1.09 | 1.07 | 0.02 | 1120 | 0.08 | 2.22 | 0.02 |
| <b>16B09_2</b> | 91  | 1.04 | 110  | 0.04 | 0.04 | 9.58 | 1.45 | 0.17 | 3.69 | 6.17 | 0.84 | 36.7 | 0.06 | 0.28 | 1.87 | 0.05 | 827  | 0.11 | 5.02 | 0.02 |
| <b>16B09_2</b> | 92  | 1.57 | 114  | 0.05 | 0.09 | 32.3 | 1.63 | 0.67 | 5.24 | 6.85 | 1.72 | 36.8 | 0.16 | 30.5 | 11.8 | 0.03 | 850  | 0.18 | 1.71 | 0.05 |
| <b>16B09_2</b> | 96  | 1.11 | 107  | 0.05 | 0.11 | 22.1 | 0.74 | 1.01 | 4.10 | 6.13 | 1.15 | 36.8 | 0.10 | 15.3 | 5.14 | 0.04 | 793  | 0.10 | 16.1 | 0.03 |
| <b>16B09_2</b> | 98  | 0.81 | 125  | 0.05 | 0.11 | 62.4 | 0.78 | 0.20 | 5.58 | 6.46 | 1.11 | 37.0 | 0.34 | 103  | 5.72 | 0.05 | 962  | 0.13 | 3.8  | 0.02 |
| <b>16B09_2</b> | 99  | 0.93 | 125  | 0.06 | 0.13 | 39.9 | 0.89 | 0.16 | 7.50 | 7.67 | 1.14 | 36.7 | 0.25 | 50.1 | 2.94 | 0.04 | 947  | 0.19 | 3.43 | 0.02 |
| <b>16B09_2</b> | 102 | 1.40 | 131  | 0.05 | 0.16 | 72.8 | 5.26 | 0.86 | 4.86 | 4.92 | 2.24 | 32.7 | 0.40 | 165  | 22.7 | 0.04 | 915  | 0.15 | 21.4 | 0.02 |
| <b>16B09_2</b> | 103 | 1.12 | 129  | 0.04 | 0.06 | 15.5 | 2.19 | 0.22 | 4.38 | 5.37 | 0.94 | 35.2 | 0.06 | 3.58 | 2.39 | 0.04 | 973  | 0.12 | 8.68 | 0.02 |
| <b>16B09_2</b> | 105 | 1.18 | 136  | 0.05 | 0.15 | 34.4 | 6.72 | 0.52 | 7.80 | 8.94 | 1.40 | 37.2 | 0.18 | 34.0 | 8.65 | 0.03 | 1169 | 0.26 | 6.00 | 0.01 |
| <b>16B09_2</b> | 107 | 0.85 | 123  | 0.04 | 0.07 | 21.7 | 8.64 | 0.37 | 6.99 | 5.66 | 1.33 | 37.1 | 0.12 | 8.21 | 12.9 | 0.03 | 1090 | 0.14 | 3.20 | 0.02 |
| <b>16B09_2</b> | 109 | 0.79 | 114  | 0.05 | 0.12 | 16.4 | 1.21 | 0.19 | 6.12 | 7.09 | 1.12 | 38.0 | 0.19 | 6.54 | 3.70 | 0.02 | 945  | 0.17 | 4.59 | 0.02 |

| QCM            |            | Mn   | Fe     | Co   | Ni   | Cu     | Ga   | Ge   | As   | Se   | Ag   | Cd   | In   | Sn   | Sb   | Au** | Hg*  | Tl   | Pb   | Bi*  |
|----------------|------------|------|--------|------|------|--------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| NIST SRM 610** | Mean       | 488  | 628    | 466  | 438  | 559    | 492  | 394  | 473  | 109  | 218  | 216  | 418  | 447  | 558  |      | 2.35 | 134  | 525  | 356  |
|                | 1 SD       | 8    | 22.3   | 8.36 | 9.57 | 12.1   | 8.87 | 10.0 | 13.9 | 13.1 | 5.29 | 5.39 | 10.6 | 13.0 | 13.4 |      | 0.57 | 5.50 | 22.9 | 11.4 |
|                | Ref. Value | 444  | 458    | 410  | 458  | 441    | 433  | 447  | 325  | 138  | 251  | 270  | 434  | 430  | 396  |      | N/A  | 59.6 | 426  | 384  |
|                | 95% CL     | 13   | 9      | 10   | 4    | 15     | 13   | 78   | 18   | 42   | 9    | 16   | 19   | 29   | 19   |      | -    | 2.8  | 1    | 26   |
| USGS MASS-1*   | Mean       | 270  | 155000 | 71.9 | 89.7 | 145000 | 64.3 | 53.6 | 77.1 | 45.7 | 52.0 | 86.7 | 55.8 | 58.2 | 84.6 | 50.8 | 201  | 136  | 87.6 | 49.4 |
|                | 1 SD       | 7.24 | 3390   | 1.84 | 2.41 | 3440   | 1.39 | 2.36 | 2.92 | 7.95 | 1.65 | 4.20 | 2.11 | 2.02 | 2.47 | 17.8 | 282  | 7.57 | 7.00 | 2.89 |
|                | Ref. Value | 260  | 156000 | 67   | N/A  | 134000 | 50   | 50   | 65   | 53   | 67   | 70   | 50   | 55   | 55   | 47   | N/A  | N/A  | 75.5 | 7    |
|                | 1 SD       | N/A  | 3900   | 6    | N/A  | 6700   | 5    | N/A  | 7    | 5    | 6    | 11   | N/A  | N/A  | 6    | 41   | N/A  | N/A  | N/A  | 1    |
| MUL-ZnS-1      | Mean       |      |        |      |      |        |      |      |      |      |      |      |      |      |      | 1.27 |      |      |      |      |
|                | 1 SD       |      |        |      |      |        |      |      |      |      |      |      |      |      |      | 1.49 |      |      |      |      |
|                | Ref. Value |      |        |      |      |        |      |      |      |      |      |      |      |      |      | N/A  |      |      |      |      |
|                | 1 SD       |      |        |      |      |        |      |      |      |      |      |      |      |      |      | N/A  |      |      |      |      |

For NIST SRM 610 the preferred values reported in the GeoReM database (<http://georem.mpch-mainz.gwdg.de>) (Jochum et al. 2005; 2011) were taken. The reference values for USGS MASS-1 were taken from Wilson et al. (2002) and the reference values for MUL-ZnS-1 were taken from Onuk et al. (2017).

**Table 2: Trace element concentrations in galena. For calibration USGS MASS-1 was used for most elements. For elements marked with \* MUL-1-ZnS, with \*\*NIST SRM 610 was applied as calibration material. All values are given in µg/g. Also given are obtained trace element concentrations for analysed quality control materials, together with respective reference values if available.**

| Sample       | Spot | Mn   | Fe   | Co  | Ni*   | Cu   | Zn  | Ga  | Ge   | As   | Se   | Sr** | Mo    | Ag*  | Cd   | In   | Sn    | Sb   | Au** | Hg   | Tl   | Bi   |
|--------------|------|------|------|-----|-------|------|-----|-----|------|------|------|------|-------|------|------|------|-------|------|------|------|------|------|
| <b>16B07</b> | 11   | 6.37 | 23.9 | bdl | bdl   | bdl  | bdl | bdl | 6.14 | 40.7 | 1030 | 0.03 | 0.308 | 38.7 | 4.47 | 0.14 | 45.7  | 55.2 | 0.01 | 3.85 | 0.52 | 1.01 |
| <b>16B07</b> | 12   | 0.48 | 5630 | bdl | 0.59  | 10.2 | bdl | bdl | 5.55 | 94.4 | 1060 | bdl  | 2.42  | 37.3 | 1.88 | bdl  | 0.630 | 43.2 | 0.05 | 8.82 | 0.41 | 1.07 |
| <b>16B07</b> | 14   | 9.84 | 24.3 | bdl | 0.105 | 2.71 | bdl | bdl | 4.03 | 49.4 | 725  | 0.02 | 2.59  | 37.6 | 2.35 | 0.01 | 0.07  | 53.6 | 0.05 | 4.82 | 0.50 | 0.05 |

| 16B07   | 16   | 7.63 | 973  | bdl   | 0.06 | 20.4 | 5.77  | bdl  | 4.09 | 50.7 | 663  | 0.01  | 4.54 | 33.5 | 2.80 | bdl   | 0.14 | 49.0 | 0.03  | 6.96 | 0.51 | 0.09 |
|---------|------|------|------|-------|------|------|-------|------|------|------|------|-------|------|------|------|-------|------|------|-------|------|------|------|
| 16B07   | 17   | 1.52 | 12.3 | 0.07  | 0.03 | 3.19 | 6.46  | bdl  | 4.03 | 50.7 | 712  | 0.02  | 0.25 | 38.6 | 2.76 | 0.01  | bdl  | 53.3 | 0.09  | 8.17 | 0.47 | 0.09 |
| 16B06   | 19   | 64.8 | 122  | bdl   | bdl  | 2.88 | bdl   | 0.20 | 5.14 | 43.7 | 903  | 0.68  | 3.26 | 50.9 | 4.34 | 0.77  | 258  | 61.8 | bdl   | 3.16 | 0.74 | 29.9 |
| 16B06   | 21   | 1.01 | bdl  | bdl   | bdl  | 2.03 | bdl   | bdl  | 5.86 | 38.8 | 1060 | 0.01  | bdl  | 29.9 | 3.88 | 0.29  | 104  | 37.7 | 0.06  | 5.35 | 0.47 | 8.57 |
| 16B06   | 22   | 4.52 | 10.5 | bdl   | bdl  | 3.01 | 1.69  | bdl  | 6.51 | 42.2 | 1040 | 1.18  | bdl  | 29.8 | 2.46 | 0.59  | 186  | 37.3 | 0.02  | 4.66 | 0.51 | 8.68 |
| 16B06   | 23   | bdl  | bdl  | bdl   | bdl  | 3.28 | bdl   | 0.02 | 6.00 | 41.9 | 975  | bdl   | bdl  | 31.3 | 4.24 | bdl   | 1.38 | 42.2 | 0.01  | 2.52 | 0.61 | 7.98 |
| 16B09_2 | 27   | 610  | 2990 | 1.14  | 2.97 | 48.4 | 230   | 0.73 | 7.39 | 40.5 | 801  | 0.82  | 24.4 | 62.6 | 39.4 | 1.66  | 122  | 128  | 0.24  | 5.49 | 1.20 | 1.85 |
| 16B09_2 | 28   | 415  | 7450 | 0.19  | bdl  | 25.5 | 6.11  | 0.06 | 4.87 | 61.4 | 866  | 0.29  | 37.6 | 55.3 | 2.81 | bdl   | 0.68 | 87.3 | 0.04  | 7.07 | 0.88 | 0.17 |
| 16B09_2 | 29   | 0.58 | bdl  | bdl   | 0.20 | 2.97 | 4.03  | 0.05 | 5.48 | 34.6 | 947  | 0.10  | bdl  | 73.4 | 5.36 | 0.01  | bdl  | 111  | bdl   | 3.11 | 1.16 | 0.13 |
| 16B09_2 | 43   | 3.24 | 1190 | bdl   | bdl  | 17.0 | 4.41  | bdl  | 5.20 | 29.2 | 763  | 22.4  | 2.60 | 50.5 | 3.78 | bdl   | 0.07 | 82.4 | bdl   | 4.59 | 0.74 | 0.07 |
| 16B09_2 | 44   | bdl  | 9.59 | bdl   | bdl  | 6.19 | bdl   | bdl  | 4.62 | 19.0 | 794  | 0.06  | bdl  | 57.6 | 3.77 | bdl   | 0.11 | 87.7 | bdl   | 3.57 | 0.80 | 0.13 |
| 16B09_2 | 45   | 2.9  | bdl  | 0.035 | bdl  | 11.2 | 2.02  | bdl  | 5.28 | 33.1 | 1010 | 0.03  | bdl  | 72.0 | 4.31 | bdl   | 0.16 | 112  | bdl   | 3.81 | 1.01 | 0.16 |
| 16B09_2 | 46   | 0.44 | 6170 | bdl   | 0.17 | 4.53 | bdl   | bdl  | 5.33 | 51.3 | 882  | 1.09  | 0.07 | 63.9 | 4.96 | 0.01  | 4.32 | 95.1 | 0.00  | 5.22 | 0.97 | 0.12 |
| 16B09_2 | 49   | 6.22 | 1670 | bdl   | 0.16 | 3.42 | bdl   | bdl  | 5.83 | 34.2 | 899  | 0.07  | 6.65 | 58.4 | 4.29 | bdl   | 0.36 | 89.6 | bdl   | 6.70 | 0.84 | 0.09 |
| 16B09_2 | 50   | 8.48 | 23.2 | bdl   | bdl  | 4.3  | bdl   | 0.14 | 5.51 | 34.5 | 913  | 0.23  | 6.68 | 60.4 | 3.44 | bdl   | 0.74 | 91.8 | bdl   | 4.36 | 0.79 | 0.09 |
|         |      |      |      |       |      |      |       |      |      |      |      |       |      |      |      |       |      |      |       |      |      |      |
| 1 SD    |      |      |      |       |      |      |       |      |      |      |      |       |      |      |      |       |      |      |       |      |      |      |
| Sample  | Spot | Mn   | Fe   | Co    | Ni*  | Cu   | Zn    | Ga   | Ge   | As   | Se   | Sr**  | Mo   | Ag*  | Cd   | In    | Sn   | Sb   | Au**  | Hg   | Tl   | Bi   |
| 16B07   | 11   | 0.44 | 4.05 | 0.02  | 0.04 | 0.39 | 0.68  | 0.02 | 0.34 | 7.14 | 169  | 0.01  | 0.07 | 1.98 | 0.60 | 0.02  | 1.72 | 2.73 | 0.01  | 0.70 | 0.04 | 0.06 |
| 16B07   | 12   | 0.20 | 277  | 0.04  | 0.30 | 1.52 | 1.55  | 0.01 | 0.64 | 16.9 | 177  | 0.003 | 0.57 | 2.29 | 0.94 | 0.002 | 0.20 | 2.51 | 0.04  | 1.65 | 0.07 | 0.12 |
| 16B07   | 14   | 0.69 | 5.38 | 0.03  | 0.09 | 0.56 | 0.91  | 0.01 | 0.33 | 8.79 | 120  | 0.01  | 0.35 | 2.01 | 0.56 | 0.01  | 0.05 | 2.73 | 0.02  | 0.88 | 0.05 | 0.02 |
| 16B07   | 16   | 0.54 | 48.9 | 0.02  | 0.04 | 1.49 | 0.92  | 0.02 | 0.30 | 8.93 | 110  | 0.01  | 0.43 | 1.78 | 0.53 | 0.003 | 0.06 | 2.48 | 0.01  | 1.23 | 0.04 | 0.02 |
| 16B07   | 17   | 0.15 | 3.90 | 0.03  | 0.03 | 0.46 | 0.94  | bdl  | 0.28 | 8.89 | 118  | 0.01  | 0.08 | 2.01 | 0.50 | 0.005 | 0.04 | 2.66 | 0.02  | 1.42 | 0.04 | 0.02 |
| 16B06   | 19   | 4.24 | 12.3 | 0.03  | 0.04 | 0.68 | 0.93  | 0.09 | 0.49 | 8.16 | 150  | 0.09  | 0.51 | 2.79 | 0.90 | 0.08  | 9.59 | 3.24 | 0.02  | 0.64 | 0.08 | 1.23 |
| 16B06   | 21   | 0.16 | 4.95 | 0.03  | 0.06 | 0.54 | 0.91  | 0.03 | 0.45 | 7.16 | 176  | 0.01  | 0.05 | 1.66 | 0.73 | 0.04  | 4.00 | 2.00 | 0.03  | 0.98 | 0.05 | 0.39 |
| 16B06   | 22   | 0.43 | 6.48 | 0.03  | 0.04 | 0.76 | 1.17  | 0.01 | 0.60 | 8.02 | 173  | 0.13  | 0.02 | 1.78 | 0.73 | 0.08  | 7.13 | 2.10 | 0.02  | 0.90 | 0.07 | 0.43 |
| 16B06   | 23   | 0.11 | 4.83 | 0.03  | 0.05 | 0.59 | 0.94  | 0.02 | 0.45 | 7.64 | 161  | 0.004 | bdl  | 1.73 | 0.76 | 0.002 | 0.18 | 2.20 | 0.01  | 0.51 | 0.06 | 0.36 |
| 16B09_2 | 27   | 38.9 | 144  | 0.12  | 0.63 | 3.20 | 14.39 | 0.12 | 0.49 | 7.29 | 133  | 0.08  | 1.51 | 3.26 | 3.88 | 0.10  | 4.57 | 6.34 | 0.05  | 0.99 | 0.08 | 0.11 |
| 16B09_2 | 28   | 26.5 | 358  | 0.07  | 0.08 | 2.19 | 1.36  | 0.05 | 0.49 | 11.1 | 144  | 0.06  | 2.47 | 3.04 | 0.78 | 0.002 | 0.16 | 4.50 | 0.03  | 1.29 | 0.09 | 0.04 |
| 16B09_2 | 29   | 0.15 | 5.81 | 0.02  | 0.14 | 0.67 | 1.18  | 0.04 | 0.48 | 6.57 | 157  | 0.03  | 0.02 | 3.89 | 0.94 | 0.01  | 0.06 | 5.60 | 0.005 | 0.61 | 0.10 | 0.03 |
| 16B09_2 | 43   | 0.40 | 69.7 | 0.04  | 0.02 | 2.03 | 1.65  | 0.07 | 0.66 | 6.24 | 129  | 1.06  | 0.62 | 3.01 | 1.09 | 0.003 | 0.09 | 4.48 | bdl   | 1.00 | 0.10 | 0.04 |
| 16B09_2 | 44   | 0.10 | 4.92 | 0.03  | 0.08 | 0.73 | 0.81  | 0.02 | 0.34 | 3.84 | 131  | 0.02  | bdl  | 2.99 | 0.66 | 0.002 | 0.05 | 4.36 | bdl   | 0.67 | 0.06 | 0.02 |
| 16B09_2 | 45   | 0.25 | 4.64 | 0.02  | 0.05 | 0.95 | 0.82  | 0.02 | 0.36 | 6.02 | 167  | 0.01  | 0.02 | 3.69 | 0.65 | 0.002 | 0.06 | 5.52 | 0.01  | 0.69 | 0.07 | 0.02 |

|         |    |      |      |      |      |      |      |      |      |      |     |      |      |      |      |       |      |      |       |      |      |      |
|---------|----|------|------|------|------|------|------|------|------|------|-----|------|------|------|------|-------|------|------|-------|------|------|------|
| 16B09_2 | 46 | 0.09 | 291  | 0.02 | 0.08 | 0.48 | 0.60 | 0.01 | 0.30 | 8.90 | 145 | 0.06 | 0.04 | 3.23 | 0.61 | 0.005 | 0.23 | 4.64 | 0.004 | 0.91 | 0.05 | 0.02 |
| 16B09_2 | 49 | 0.54 | 87.7 | 0.05 | 0.13 | 0.81 | 1.15 | 0.01 | 0.58 | 6.82 | 150 | 0.03 | 0.84 | 3.23 | 0.98 | 0.002 | 0.13 | 4.65 | 0.02  | 1.24 | 0.09 | 0.03 |
| 16B09_2 | 50 | 0.61 | 5.73 | 0.03 | 0.02 | 0.59 | 0.78 | 0.06 | 0.40 | 6.31 | 151 | 0.04 | 0.60 | 3.15 | 0.61 | 0.003 | 0.12 | 4.57 | 0.01  | 0.79 | 0.06 | 0.02 |

| QCM            |            | Mn   | Fe    | Co   | Ni*  | Cu   | Zn     | Ga   | Ge   | As   | Se   | Sr** | Mo   | Ag*  | Cd    | In   | Sn   | Sb   | Au** | Hg  | Tl   | Bi   |
|----------------|------------|------|-------|------|------|------|--------|------|------|------|------|------|------|------|-------|------|------|------|------|-----|------|------|
| NIST SRM 610** | Mean       | 482  | 580   | 375  | 386  | 482  | 469    | 474  | 361  | 383  | 116  |      | 471  | 167  | 207   | 367  | 443  | 388  |      | bdl | 45.8 | 348  |
|                | 1 SD       | 6.42 | 19.3  | 3.48 | 12.8 | 6.87 | 61.7   | 5.49 | 4.58 | 6.73 | 9.88 |      | 7.48 | 1.77 | 5.30  | 5.71 | 6.79 | 2.93 |      | bdl | 0.83 | 2.78 |
|                | Ref. Value | 444  | 458   | 410  | 458  | 441  | 460    | 433  | 447  | 325  | 138  |      | 417  | 251  | 270   | 434  | 430  | 396  |      | N/A | 59.6 | 384  |
|                | 95% CL     | 13   | 9     | 10   | 4    | 15   | 18     | 13   | 78   | 18   | 42   |      | 21   | 9    | 16    | 19   | 29   | 19   |      |     | 2.8  | 26   |
| MUL-ZnS-1*     | Mean       | 785  | 66500 | 319  |      | 1098 | 723000 | 714  | 569  | 76.9 | 273  | 165  | 88.1 |      | 13163 | 572  | 1380 | 760  | 2.04 | 205 | 4.98 | 61.7 |
|                | 1 SD       | 45.4 | 7200  | 37.1 |      | 28.8 | 17600  | 156  | 106  | 8.23 | 8.02 | 4.89 | 6.61 |      | 530   | 6.77 | 55.2 | 18.7 | 1.22 | 102 | 0.19 | 1.03 |
|                | Ref. Value | 611  | 55800 | 308  |      | 994  | 585500 | 602  | 505  | 65.9 | 200  | N/A  | 59.8 |      | 11400 | 515  | 1048 | 819  | N/A  | N/A | 11.1 | N/A  |
|                | 1 SD       | 42.7 | 1400  | 7.70 |      | 25.9 | 14600  | 13.8 | 32.3 | 3.30 | 2.00 | N/A  | 5.39 |      | 200   | 13.9 | 83.9 | 10.7 | N/A  | N/A | 0.36 | N/A  |
| USGS MASS-1    | Mean       |      |       |      | 78.8 |      |        |      |      |      |      | 0.84 |      | 40.3 |       |      |      |      | 52.6 |     |      |      |
|                | 1 SD       |      |       |      | 9.80 |      |        |      |      |      |      | 0.89 |      | 4.72 |       |      |      |      | 6.26 |     |      |      |
|                | Ref. Value |      |       |      | N/A  |      |        |      |      |      |      | N/A  |      | 67.0 |       |      |      |      | 47   |     |      |      |
|                | 1 SD       |      |       |      | N/A  |      |        |      |      |      |      | N/A  |      | 6    |       |      |      |      | 41   |     |      |      |

For NIST SRM 610 the preferred values reported in the GeoReM database (<http://georem.mpch-mainz.gwdg.de>) (Jochum et al. 2005; 2011) were taken. The reference values for USGS MASS-1 were taken from Wilson et al. (2002) and the reference values for MUL-ZnS-1 were taken from Onuk et al. (2017).

**Table 3: Trace element concentrations in pyrite. For calibration USGS MASS-1 was used for most elements. For elements marked with \* MUL-1-ZnS, with \*\*NIST SRM 610 was applied as calibration material. All values are given in µg/g. Also given are obtained trace element concentrations for analysed quality control materials, together with respective reference values if available.**

| Sample | Spot | Mn   | Co   | Ni*  | Cu   | Zn   | Ga   | Ge   | As    | Se   | Sr** | Mo   | Ag*  | Cd   | In   | Sn   | Sb   | Au** | Hg   | Tl*  | Pb*  | Bi   |
|--------|------|------|------|------|------|------|------|------|-------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 16B07  | 10   | 0.23 | 0.27 | 1.85 | 91.9 | 10.1 | 0.09 | 0.74 | 6.27  | 57.8 | bdl  | 0.14 | 0.22 | 1.15 | 0.01 | 0.09 | 1.31 | bdl  | 1.57 | 0.04 | 1.14 | 0.02 |
| 16B07  | 11   | 0.32 | 0.46 | 2.17 | 166  | 10.8 | 0.10 | 1.06 | 6.92  | 71.5 | bdl  | 0.34 | 0.05 | 1.62 | 0.01 | 0.15 | 0.33 | bdl  | 0.83 | 0.05 | 0.31 | 0.03 |
| 16B07  | 14   | 1000 | 0.06 | 1.56 | 746  | 76.1 | 0.30 | 0.43 | 8075  | 3.23 | bdl  | 14.9 | 2.42 | 0.96 | 0.01 | 0.64 | 494  | 2.17 | 229  | 34.0 | 97.6 | 0.03 |
| 16B07  | 15   | 1350 | 0.06 | 1.81 | 562  | 72.9 | 0.37 | 0.37 | 6805  | 2.42 | bdl  | 23.7 | 2.2  | 0.81 | 0.01 | 0.62 | 366  | 1.83 | 189  | 38.1 | 182  | 0.02 |
| 16B07  | 16   | 9.65 | 0.06 | 0.52 | 5100 | 43.1 | 0.08 | 0.37 | 500   | 64.3 | 6.09 | 4.85 | 3.34 | 0.85 | 0.01 | 0.59 | 214  | 0.57 | 29.8 | 0.22 | 276  | 0.02 |
| 16B07  | 17   | 3.73 | 2.52 | 0.66 | 6680 | 14.6 | 0.09 | 1.63 | 144   | 33.4 | bdl  | 0.15 | 4.39 | 0.94 | 0.01 | 0.14 | 37.0 | 0.71 | 27.8 | 0.25 | 810  | 0.03 |
| 16B07  | 18   | 9.05 | 0.08 | 1.09 | 2020 | 8.20 | 0.06 | 0.25 | 268   | 20   | bdl  | 1.71 | 2.58 | 1.02 | 0.01 | 0.45 | 40.8 | 0.78 | 20.3 | 1.01 | 290  | 0.04 |
| 16B07  | 20   | 17.5 | 0.18 | 3.49 | 870  | 38.4 | 0.15 | 0.35 | 1603  | 2.68 | bdl  | 33.4 | 4.6  | 0.73 | 0.01 | 0.18 | 385  | 4.96 | 29.7 | 3.76 | 136  | 0.02 |
| 16B07  | 21   | 25.6 | 0.09 | 3.82 | 668  | 13.6 | 0.09 | 0.5  | 352   | 5.67 | bdl  | 0.1  | 3.9  | 0.86 | bdl  | 0.12 | 34.0 | 3.44 | 20.9 | 0.19 | 42.7 | 0.03 |
| 16B07  | 23   | 15.1 | 0.08 | 2.38 | 536  | 11.8 | 0.06 | 0.45 | 345   | 8.55 | bdl  | 0.28 | 3.54 | 0.74 | 0.01 | 0.09 | 44.9 | 4.00 | 17.7 | 0.15 | 35.0 | 0.02 |
| 16B07  | 24   | 365  | 0.13 | 1.12 | 321  | 30.4 | 0.03 | 0.12 | 11700 | 2.35 | bdl  | 1.91 | 2.25 | 0.50 | bdl  | 0.12 | 294  | 1.83 | 228  | 40.4 | 41.8 | 0.02 |
| 16B07  | 26   | 13.5 | 0.07 | 2.98 | 526  | 9.3  | 0.05 | 0.35 | 286   | 7.9  | bdl  | 0.14 | 3.34 | 0.77 | 0.01 | 0.09 | 42.1 | 3.91 | 15.8 | 0.16 | 45.7 | 0.02 |
| 16B07  | 28   | 21.2 | 0.07 | 2.08 | 400  | 19.3 | 0.04 | 0.16 | 197   | 7.58 | bdl  | 0    | 2.46 | 0.61 | 0.01 | 0.10 | 18.6 | 2.27 | 11.1 | 0.22 | 26.0 | 0.02 |
| 16B07  | 29   | 18.6 | 0.11 | 0.98 | 781  | 16.1 | 0.28 | 0.45 | 837   | 8.1  | bdl  | 12.6 | 5.03 | 1.24 | 0.00 | 0.49 | 122  | 4.08 | 30.2 | 0.37 | 115  | 0.03 |
| 16B07  | 30   | 350  | 0.07 | 0.53 | 450  | 31.3 | 0.05 | 0.26 | 17200 | 2.95 | bdl  | 0.08 | 3.55 | 0.76 | 0.01 | 0.11 | 706  | 3.81 | 147  | 49.9 | 62.7 | 0.03 |

| 16B06   | 31   | 9.28 | 2.55 | 0.74 | 848   | 26.5  | 0.06 | 0.25 | 472   | 2.81 | bdl   | 1.7  | 2.75 | 0.80 | 0.06  | 15.5 | 1600 | 0.75 | 61.3 | 1.98 | 39.5 | 0.03 |
|---------|------|------|------|------|-------|-------|------|------|-------|------|-------|------|------|------|-------|------|------|------|------|------|------|------|
| 16B06   | 33   | 91.0 | 1.47 | 0.96 | 460   | 106   | 0.11 | 1.11 | 681   | 7.18 | bdl   | 0.65 | 1.88 | 1.08 | 0.01  | 1.39 | 81.2 | 0.77 | 20.0 | 2.22 | 28.5 | 0.06 |
| 16B06   | 34   | 700  | 18.7 | 13.2 | 1410  | 371   | 0.56 | 0.76 | 3820  | 12.4 | bdl   | 120  | 5.17 | 0.99 | 0.02  | 4.65 | 1080 | 1.94 | 79.3 | 18   | 128  | 0.16 |
| 16B06   | 35   | 28.9 | 1.31 | 1.42 | 546   | 48.3  | 0.09 | 0.41 | 694   | 6.32 | bdl   | 1.49 | 1.53 | 0.54 | 0.02  | 2.33 | 168  | 1.06 | 6.19 | 1.75 | 36.7 | 0.02 |
| 16B06   | 36   | 22.8 | 0.07 | 1.26 | 328   | 18.4  | 0.05 | 0.74 | 4960  | 3.29 | bdl   | 0.46 | 1.17 | 0.59 | 0.01  | 1.31 | 15.6 | 0.76 | 68.1 | 63.4 | 5.35 | 0.02 |
| 16B06   | 38   | 14.8 | 13.3 | 0.1  | 172   | 8.60  | 0.05 | 0.36 | 1280  | 13.5 | bdl   | 3.18 | 0.13 | 0.71 | 0.01  | 0.14 | 70.3 | 0.09 | 41.9 | 5.09 | 4.5  | 0.03 |
| 16B06   | 51   | 4.35 | 5.89 | 0.29 | 83.5  | 9.75  | 0.10 | 0.28 | 319   | 9.61 | bdl   | 0.61 | 0.25 | 1.13 | 0.01  | 0.57 | 37.4 | bdl  | 5.03 | 0.67 | 10.6 | 0.02 |
| 16B09_2 | 52   | 3.92 | 0.14 | 0.25 | 257   | 5.16  | 0.03 | 0.20 | 127   | 4.79 | bdl   | 0.7  | 1.32 | 0.68 | 0.01  | 0.09 | 27.7 | 0.75 | 4.29 | 0.22 | 58.2 | 0.02 |
| 16B09_2 | 53   | 3.16 | 0.12 | 0.21 | 161   | 9.29  | 0.11 | 0.31 | 238   | 5.28 | bdl   | 0.22 | 1.04 | 1.14 | 0.02  | 0.20 | 17.3 | 1.34 | 7.97 | 1.7  | 41.5 | 0.04 |
| 16B09_2 | 54   | 2.12 | 0.08 | 0.18 | 1630  | 13.5  | 0.09 | 0.51 | 89.8  | 4.7  | bdl   | 4.46 | 0.89 | 1.10 | 0.02  | 0.10 | 18.2 | 0.24 | 5.49 | 0.39 | 249  | 0.04 |
| 16B09_2 | 58   | 57.7 | 0.06 | 0.22 | 861   | 95.8  | 0.06 | 2.00 | 2100  | 2.22 | bdl   | 0.97 | 4.05 | 0.60 | 0.01  | 0.17 | 3349 | 0.86 | 53.4 | 1.42 | 247  | 0.01 |
| 16B09_2 | 59   | 24.6 | 0.05 | 0.10 | 140   | 21.5  | 0.03 | 3.57 | 1210  | 2.93 | bdl   | 0.58 | 0.86 | 0.66 | 0.01  | 0.08 | 28.2 | 0.57 | 22.1 | 49.0 | 6.84 | 0.03 |
| 16B09_2 | 60   | 55.9 | 0.08 | 0.98 | 1720  | 26.6  | 0.04 | 0.66 | 2880  | 79.3 | 12.8  | 41.1 | 26.5 | 0.90 | 0.01  | 0.30 | 756  | 9.1  | 157  | 1.63 | 5690 | 0.03 |
| 16B09_2 | 61   | 44.0 | 2.70 | 0.83 | 2750  | 791   | 0.06 | 0.35 | 2130  | 50.9 | bdl   | 6.98 | 9.98 | 1.74 | 0.01  | 0.13 | 2430 | 2.58 | 133  | 1.14 | 1690 | 0.03 |
| 16B09_2 | 62   | 30.7 | 2.48 | 0.41 | 984   | 66.8  | bdl  | 0.17 | 1230  | 33.2 | bdl   | 17.6 | 3.85 | 0.57 | 0.01  | 1.73 | 1970 | 1.42 | 25.8 | 0.34 | 314  | 0.02 |
| 16B09_2 | 65   | 29.5 | 1.02 | 1.45 | 1370  | 244   | 0.07 | 0.26 | 1560  | 13.5 | bdl   | 5.29 | 7.18 | 0.86 | 0.01  | 0.18 | 2870 | 2.89 | 92.0 | 0.77 | 130  | 0.03 |
| 16B09_2 | 66   | 213  | 0.05 | 2.03 | 3930  | 6320  | 0.71 | 0.36 | 9400  | 2.58 | bdl   | 57.1 | 8.49 | 6.79 | 0.01  | 1.78 | 1460 | 4.82 | 106  | 8.84 | 97.9 | 0.01 |
| 16B09_2 | 67   | 509  | 0.06 | 0.93 | 22900 | 16100 | 1.52 | 0.63 | 25900 | 3.26 | bdl   | 161  | 37.5 | 30.4 | 0.04  | 3.17 | 3330 | 27.6 | 619  | 70.3 | 653  | 0.02 |
| 16B09_2 | 68   | 455  | 0.64 | 2.83 | 632   | 24.7  | 0.11 | 0.88 | 1670  | 11.2 | bdl   | 34.0 | 2.31 | 0.93 | 0.02  | 0.29 | 156  | 1.06 | 23.1 | 6.02 | 54.2 | 0.04 |
| 16B09_2 | 69   | 12.6 | 0.05 | 0.88 | 596   | 22.4  | 0.07 | 0.12 | 772   | 17.9 | bdl   | 7.86 | 3.6  | 1.08 | 0.01  | 0.31 | 1530 | 1.15 | 24.0 | 0.44 | 30.9 | 0.01 |
| 16B09_2 | 70   | 3.49 | 0.04 | 0.47 | 474   | 25.7  | 0.08 | 0.33 | 794   | 12.5 | bdl   | 11.7 | 2.55 | 0.42 | 0.02  | 0.07 | 1650 | 0.96 | 10.4 | 0.28 | 44.3 | 0.02 |
|         |      |      |      |      |       |       |      |      |       |      |       |      |      |      |       |      |      |      |      |      |      |      |
| 1 SD    |      |      |      |      |       |       |      |      |       |      |       |      |      |      |       |      |      |      |      |      |      |      |
| Sample  | Spot | Mn   | Co   | Ni*  | Cu    | Zn    | Ga   | Ge   | As    | Se   | Sr*** | Mo   | Ag*  | Cd   | In    | Sn   | Sb   | Au** | Hg   | Tl*  | Pb*  | Bi   |
| 16B07   | 10   | 0.14 | 0.07 | 1.13 | 4.42  | 6.00  | 0.05 | 0.21 | 3.04  | 7.59 | 2.56  | 0.10 | 0.19 | 0.68 | 0.01  | 0.07 | 0.25 | bdl  | 0.73 | 0.02 | 1.03 | 0.01 |
| 16B07   | 11   | 0.16 | 0.08 | 1.15 | 7.55  | 5.44  | 0.03 | 0.20 | 3.34  | 7.86 | 2.98  | 0.13 | 0.05 | 0.77 | 0.003 | 0.08 | 0.17 | bdl  | 0.46 | 0.02 | 0.27 | 0.02 |
| 16B07   | 14   | 37.4 | 0.03 | 0.73 | 32.8  | 5.89  | 0.06 | 0.10 | 792   | 1.43 | 2.75  | 0.88 | 1.29 | 0.41 | 0.004 | 0.09 | 22.4 | 1.07 | 96.0 | 10.1 | 90.2 | 0.01 |
| 16B07   | 15   | 50.3 | 0.03 | 0.93 | 24.8  | 6.42  | 0.08 | 0.11 | 667   | 1.43 | 1.84  | 1.41 | 1.22 | 0.41 | 0.01  | 0.11 | 16.7 | 0.91 | 79.4 | 11.3 | 168  | 0.01 |
| 16B07   | 16   | 0.45 | 0.03 | 0.37 | 223   | 5.29  | 0.03 | 0.11 | 49.3  | 6.66 | 3.74  | 0.46 | 1.79 | 0.42 | 0.005 | 0.10 | 9.79 | 0.29 | 12.5 | 0.07 | 255  | 0.01 |
| 16B07   | 17   | 0.23 | 0.16 | 0.43 | 292   | 3.86  | 0.04 | 0.19 | 14.6  | 4.11 | 2.12  | 0.08 | 2.32 | 0.41 | 0.01  | 0.07 | 1.79 | 0.36 | 11.7 | 0.08 | 748  | 0.01 |
| 16B07   | 18   | 0.39 | 0.03 | 0.51 | 88.2  | 3.28  | 0.03 | 0.10 | 26.5  | 2.57 | 2.03  | 0.19 | 1.35 | 0.39 | 0.01  | 0.07 | 1.9  | 0.39 | 8.5  | 0.3  | 268  | 0.01 |
| 16B07   | 20   | 0.74 | 0.05 | 1.49 | 38.2  | 4.85  | 0.05 | 0.09 | 157   | 1.47 | 1.83  | 1.84 | 2.44 | 0.36 | 0.004 | 0.06 | 17.5 | 2.44 | 12.4 | 1.12 | 126  | 0.01 |
| 16B07   | 21   | 1.09 | 0.04 | 1.74 | 29.5  | 4.73  | 0.04 | 0.14 | 35.0  | 2.62 | 3.02  | 0.07 | 2.12 | 0.52 | bdl   | 0.08 | 1.73 | 1.71 | 8.76 | 0.06 | 39.4 | 0.01 |

|         |    |      |      |      |      |      |      |      |      |      |      |      |      |      |       |      |      |      |      |       |      |       |
|---------|----|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|------|------|------|------|-------|------|-------|
| 16B07   | 23 | 0.63 | 0.03 | 1.03 | 23.6 | 3.03 | 0.04 | 0.10 | 34.0 | 1.94 | 1.75 | 0.10 | 1.87 | 0.33 | 0.004 | 0.04 | 2.13 | 1.97 | 7.41 | 0.05  | 32.4 | 0.01  |
| 16B07   | 24 | 13.7 | 0.05 | 0.71 | 14.3 | 4.78 | 0.03 | 0.09 | 1150 | 1.63 | 1.89 | 0.31 | 1.26 | 0.31 | bdl   | 0.06 | 13.5 | 0.92 | 95.6 | 12.0  | 38.6 | 0.01  |
| 16B07   | 26 | 0.56 | 0.03 | 1.18 | 23.1 | 2.80 | 0.02 | 0.10 | 28.2 | 1.81 | 2.05 | 0.07 | 1.76 | 0.33 | 0.004 | 0.04 | 1.98 | 1.92 | 6.65 | 0.05  | 42.2 | 0.01  |
| 16B07   | 28 | 0.97 | 0.05 | 1.25 | 17.9 | 5.01 | 0.04 | 0.11 | 20.0 | 2.66 | 2.26 | bdl  | 1.43 | 0.36 | 0.003 | 0.06 | 1.1  | 1.14 | 4.67 | 0.08  | 24.0 | 0.01  |
| 16B07   | 29 | 0.82 | 0.05 | 0.64 | 34.4 | 5.23 | 0.08 | 0.14 | 82.4 | 2.69 | 2.97 | 0.94 | 2.68 | 0.55 | 0.005 | 0.11 | 5.7  | 2.01 | 12.7 | 0.12  | 106  | 0.02  |
| 16B07   | 30 | 13.1 | 0.03 | 0.4  | 19.9 | 5.17 | 0.03 | 0.12 | 1680 | 1.46 | 2.19 | 0.05 | 1.92 | 0.39 | 0.003 | 0.06 | 32.0 | 1.88 | 61.5 | 14.79 | 57.9 | 0.01  |
| 16B06   | 31 | 0.39 | 0.14 | 0.39 | 37.1 | 3.23 | 0.03 | 0.07 | 46.4 | 1.22 | 1.68 | 0.18 | 1.44 | 0.32 | 0.01  | 0.81 | 72.2 | 0.37 | 25.7 | 0.59  | 36.5 | 0.01  |
| 16B06   | 33 | 3.54 | 0.15 | 0.67 | 20.5 | 9.56 | 0.07 | 0.23 | 67.3 | 2.97 | 3.22 | 0.21 | 1.10 | 0.57 | 0.01  | 0.21 | 3.89 | 0.40 | 8.39 | 0.67  | 26.4 | 0.02  |
| 16B06   | 34 | 26.2 | 0.89 | 4.98 | 62.1 | 21.4 | 0.14 | 0.20 | 375  | 3.52 | 5.02 | 6.07 | 2.81 | 0.49 | 0.01  | 0.44 | 49.0 | 0.98 | 33.3 | 5.35  | 119  | 0.04  |
| 16B06   | 35 | 1.29 | 0.17 | 0.99 | 24.4 | 7.55 | 0.07 | 0.15 | 68.7 | 2.78 | 2.5  | 0.36 | 0.97 | 0.43 | 0.01  | 0.32 | 7.91 | 0.55 | 2.64 | 0.54  | 33.8 | 0.02  |
| 16B06   | 36 | 0.94 | 0.03 | 0.71 | 14.5 | 3.76 | 0.03 | 0.15 | 486  | 1.62 | 2.58 | 0.13 | 0.68 | 0.32 | 0.01  | 0.16 | 0.84 | 0.39 | 28.5 | 18.8  | 4.93 | 0.01  |
| 16B06   | 38 | 0.61 | 0.57 | 0.09 | 7.62 | 2.48 | 0.02 | 0.09 | 126  | 2.36 | 2.34 | 0.29 | 0.10 | 0.29 | 0.004 | 0.05 | 3.26 | 0.05 | 17.6 | 1.51  | 4.15 | 0.01  |
| 16B06   | 51 | 0.32 | 0.35 | 0.26 | 4.00 | 4.88 | 0.04 | 0.15 | 31.9 | 3.21 | 3.58 | 0.22 | 0.20 | 0.53 | 0.01  | 0.14 | 1.91 | 0.05 | 2.14 | 0.21  | 9.78 | 0.01  |
| 16B09_2 | 52 | 0.33 | 0.07 | 0.23 | 11.7 | 3.84 | 0.01 | 0.15 | 13.4 | 2.28 | 2.59 | 0.25 | 0.86 | 0.57 | 0.003 | 0.05 | 1.57 | 0.40 | 1.84 | 0.08  | 53.8 | 0.01  |
| 16B09_2 | 53 | 0.23 | 0.04 | 0.18 | 7.22 | 3.68 | 0.04 | 0.12 | 23.8 | 2.10 | 2.65 | 0.10 | 0.60 | 0.46 | 0.01  | 0.07 | 0.9  | 0.67 | 3.36 | 0.51  | 38.4 | 0.02  |
| 16B09_2 | 54 | 0.17 | 0.03 | 0.15 | 71.1 | 3.05 | 0.03 | 0.11 | 9.25 | 1.86 | 2.49 | 0.34 | 0.50 | 0.44 | 0.005 | bdl  | 0.9  | 0.12 | 2.31 | 0.12  | 230  | 0.01  |
| 16B09_2 | 58 | 2.34 | 0.03 | 0.2  | 38.1 | 9.58 | 0.04 | 0.30 | 207  | 1.80 | 2.64 | 0.27 | 2.26 | 0.49 | 0.002 | 0.08 | 151  | 0.45 | 22.4 | 0.44  | 228  | 0.00  |
| 16B09_2 | 59 | 0.96 | 0.02 | 0.09 | 6.22 | 2.97 | 0.02 | 0.25 | 119  | 1.20 | 2.34 | 0.11 | 0.48 | 0.27 | 0.005 | 0.03 | 1.34 | 0.28 | 9.28 | 14.5  | 6.31 | 0.01  |
| 16B09_2 | 60 | 2.12 | 0.03 | 0.46 | 75.4 | 3.56 | 0.02 | 0.11 | 283  | 6.92 | 6.87 | 1.99 | 13.6 | 0.35 | 0.004 | 0.05 | 34.1 | 4.46 | 65.6 | 0.49  | 5260 | 0.01  |
| 16B09_2 | 61 | 1.7  | 0.16 | 0.48 | 120  | 32.4 | 0.03 | 0.11 | 209  | 5.22 | 2.46 | 0.53 | 5.15 | 0.54 | 0.004 | 0.05 | 110  | 1.27 | 55.7 | 0.35  | 1570 | 0.01  |
| 16B09_2 | 62 | 1.33 | 0.22 | 0.36 | 43.5 | 7.87 | bdl  | 0.13 | 121  | 5.40 | 2.37 | 1.38 | 2.15 | 0.41 | 0.01  | 0.26 | 89.2 | 0.72 | 10.9 | 0.11  | 290  | 0.01  |
| 16B09_2 | 65 | 1.18 | 0.10 | 0.74 | 60.0 | 12.8 | 0.03 | 0.11 | 154  | 2.57 | 2.43 | 0.46 | 3.74 | 0.37 | 0.004 | 0.06 | 130  | 1.43 | 38.6 | 0.24  | 120  | 0.01  |
| 16B09_2 | 66 | 8.1  | 0.03 | 1.22 | 172  | 242  | 0.15 | 0.16 | 923  | 1.64 | 2.44 | 3.25 | 4.51 | 1.76 | 0.01  | 0.25 | 65.9 | 2.39 | 44.4 | 2.64  | 90.4 | 0.01  |
| 16B09_2 | 67 | 19.1 | 0.04 | 0.66 | 1000 | 601  | 0.21 | 0.16 | 2540 | 2.21 | 2.49 | 7.84 | 19.2 | 7.03 | 0.01  | 0.33 | 151  | 13.5 | 259  | 20.8  | 603  | 0.01  |
| 16B09_2 | 68 | 17.1 | 0.11 | 1.46 | 28.0 | 6.30 | 0.06 | 0.18 | 164  | 3.37 | 3.88 | 2.08 | 1.33 | 0.54 | 0.01  | 0.09 | 7.29 | 0.54 | 9.71 | 1.8   | 50.1 | 0.02  |
| 16B09_2 | 69 | 0.92 | 0.01 | 0.8  | 27.4 | 8.26 | 0.10 | 0.16 | 77.4 | 6.41 | 3.79 | 1.37 | 2.32 | 0.81 | 0.002 | 0.20 | 69.8 | 0.65 | 10.2 | 0.16  | 28.4 | 0.004 |
| 16B09_2 | 70 | 0.30 | 0.04 | 0.41 | 21.2 | 5.39 | 0.06 | 0.15 | 78.4 | 3.41 | 2.75 | 1.08 | 1.49 | 0.23 | 0.01  | 0.05 | 74.8 | 0.50 | 4.41 | 0.09  | 40.9 | 0.01  |

| QCM            |            | Mn   | Co   | Ni* | Cu   | Zn     | Ga   | Ge   | As   | Se   | Sr** | Mo   | Ag* | Cd   | In   | Sn   | Sb   | Te   | Au** | Hg    | Tl   | Pb*  | Bi   |
|----------------|------------|------|------|-----|------|--------|------|------|------|------|------|------|-----|------|------|------|------|------|------|-------|------|------|------|
| NIST SRM 610** | Mean       | 420  | 325  | 453 | 416  | 381    | 406  | 312  | 337  | 101  |      | 397  | 195 | 155  | 317  | 382  | 338  | 204  |      | 2.1   | 39.9 | 467  | 299  |
|                | 1 SD       | 100  | 77.4 | 8.3 | 100  | 91     | 96.5 | 74.1 | 79.2 | 23.5 |      | 93.2 | 4.4 | 37.6 | 75.1 | 90.1 | 79.6 | 49.2 |      | 2.2   | 9.47 | 24.9 | 71.0 |
|                | Ref. Value | 444  | 410  | 458 | 441  | 460    | 433  | 447  | 325  | 138  |      | 417  | 251 | 270  | 434  | 430  | 396  | 302  |      | N/A   | 59.6 | 426  | 384  |
|                | 95% CL     | 13   | 10   | 4   | 15   | 18     | 13   | 78   | 18   | 42   |      | 21   | 9   | 16   | 19   | 29   | 19   | N/A  |      | N/A   | 2.8  | 1    | 26   |
| MUL-ZnS-1*     | Mean       | 611  | 252  |     | 909  | 591000 | 554  | 453  | 64.9 | 214  | 274  | 73.6 |     | 9570 | 476  | 1120 | 628  | 0.93 | 2.7  | 257.2 | 4.14 |      | 51.0 |
|                | 1 SD       | 37.4 | 23.7 |     | 62.3 | 46000  | 26.3 | 27.9 | 8.0  | 19.8 | 18.4 | 2.8  |     | 1140 | 24.6 | 86.1 | 41.1 | 0.40 | 1.4  | 316.5 | 0.31 |      | 2.84 |

|             |                   |      |      |      |      |        |      |      |      |      |      |      |      |       |      |      |      |  |      |     |      |      |     |
|-------------|-------------------|------|------|------|------|--------|------|------|------|------|------|------|------|-------|------|------|------|--|------|-----|------|------|-----|
|             | <b>Ref. Value</b> | 611  | 308  |      | 994  | 585500 | 602  | 505  | 65.9 | 200  | N/A  | 59.8 |      | 11400 | 515  | 1048 | 819  |  | N/A  | N/A | 11.1 |      | N/A |
|             | <b>1 SD</b>       | 42.7 | 7.70 |      | 25.9 | 14600  | 13.8 | 32.3 | 3.30 | 2.00 | N/A  | 5.39 |      | 200   | 13.9 | 83.9 | 10.7 |  | N/A  | N/A | 0.36 |      | N/A |
| USGS MASS-1 | <b>Mean</b>       |      |      | 103  |      |        |      |      |      |      | 0.91 |      | 52.2 |       |      |      |      |  | 86.1 |     |      | 85.1 |     |
|             | <b>1 SD</b>       |      |      | 1.23 |      |        |      |      |      |      | 0.03 |      | 1.40 |       |      |      |      |  | 16.0 |     |      | 5.16 |     |
|             | <b>Ref. Value</b> |      |      | N/A  |      |        |      |      |      |      | N/A  |      | 67.0 |       |      |      |      |  | 47.0 |     |      | 75.5 |     |
|             | <b>1 SD</b>       |      |      | N/A  |      |        |      |      |      |      | N/A  |      | 6    |       |      |      |      |  | 41   |     |      | N/A  |     |

For NIST SRM 610 the preferred values reported in the GeoReM database (<http://georem.mpch-mainz.gwdg.de>) (Jochum et al. 2005; 2011) were taken. The reference values for USGS MASS-1 were taken from Wilson et al. (2002) and the reference values for MUL-ZnS-1 were taken from Onuk et al. (2017).

**Table 4: Trace element concentrations in tetrahedrite. For calibration USGS MASS-1 was used for most elements. For elements marked with \* MUL-1-ZnS, with \*\*NIST SRM 610 was used. Elements marked with \*\*\* are retrieved from microprobe analysis. For those elements the median was taken (n=13) and is given in wt%. All other values are given in µg/g. Also given are obtained trace element concentrations for analysed quality control materials, together with respective reference values if available.**

| Sample  | Spot | S*** | Mn   | Fe*** | Co   | Ni   | Cu*** | Zn*** | Ga   | Ge   | As*** | Se   | Sr   | Mo   | Ag   | Cd   | In   | Sn   | Sb*** | Au** | Hg    | Tl   | Pb*  | Bi   |
|---------|------|------|------|-------|------|------|-------|-------|------|------|-------|------|------|------|------|------|------|------|-------|------|-------|------|------|------|
| 16B09_2 | 73   | 26.3 | 2.2  | 1.64  | bdl  | bdl  | 39.9  | 6.13  | 0.06 | 0.25 | 10.7  | 19.2 | bdl  | bdl  | 3964 | 370  | 0.03 | bdl  | 14.0  | bdl  | 11200 | 0.22 | 12.2 | bdl  |
| 16B09_2 | 74   | 26.3 | 3.2  | 1.64  | bdl  | bdl  | 39.9  | 6.13  | bdl  | bdl  | 10.7  | 14.6 | bdl  | bdl  | 3891 | 313  | bdl  | bdl  | 14.0  | bdl  | 9680  | 0.26 | 11.4 | bdl  |
| 16B09_2 | 75   | 26.3 | 3.4  | 1.64  | bdl  | 0.25 | 39.9  | 6.13  | bdl  | bdl  | 10.7  | 19.5 | bdl  | 0.10 | 3925 | 331  | 0.07 | bdl  | 14.0  | bdl  | 9340  | 0.19 | 22.6 | bdl  |
| 16B09_2 | 76   | 26.3 | 3.5  | 1.64  | bdl  | 0.68 | 39.9  | 6.13  | 0.25 | bdl  | 10.7  | 14.6 | bdl  | bdl  | 3936 | 296  | 0.12 | 0.27 | 14.0  | 0.03 | 7210  | 0.13 | 32.9 | bdl  |
| 16B09_2 | 77   | 26.3 | 2.1  | 1.64  | 0.08 | bdl  | 39.9  | 6.13  | 0.12 | bdl  | 10.7  | 16.7 | bdl  | bdl  | 3531 | 242  | 0.08 | 0.12 | 14.0  | bdl  | 4820  | bdl  | 1.22 | bdl  |
| 16B09_2 | 78   | 26.3 | 2.4  | 1.64  | bdl  | bdl  | 39.9  | 6.13  | 0.15 | 0.22 | 10.7  | 5.13 | bdl  | bdl  | 2945 | 295  | 0.07 | 0.21 | 14.0  | 0.01 | 9530  | 0.10 | 14.9 | bdl  |
|         |      |      |      |       |      |      |       |       |      |      |       |      |      |      |      |      |      |      |       |      |       |      |      |      |
| 1 SD    |      |      |      |       |      |      |       |       |      |      |       |      |      |      |      |      |      |      |       |      |       |      |      |      |
| Sample  | Spot | S*** | Mn   | Fe*** | Co   | Ni   | Cu*** | Zn*** | Ga   | Ge   | As*** | Se   | Sr   | Mo   | Ag   | Cd   | In   | Sn   | Sb*** | Au** | Hg    | Tl   | Pb*  | Bi   |
| 16B09_2 | 73   | 0.49 | 0.30 | 0.43  | 0.05 | 0.21 | 0.55  | 0.34  | 0.06 | 0.16 | 1.34  | 4.78 | 2.20 | bdl  | 132  | 39.1 | 0.02 | 0.07 | 1.99  | bdl  | 5480  | 0.07 | 0.66 | 0.02 |
| 16B09_2 | 74   | 0.49 | 0.30 | 0.43  | 0.05 | 0.13 | 0.55  | 0.34  | 0.06 | 0.12 | 1.34  | 4.06 | 2.08 | 0.07 | 130  | 33.0 | 0.01 | 0.10 | 1.99  | 0.01 | 4720  | 0.06 | 0.58 | 0.01 |
| 16B09_2 | 75   | 0.49 | 0.28 | 0.43  | 0.05 | 0.14 | 0.55  | 0.34  | 0.04 | 0.12 | 1.34  | 3.89 | 1.78 | 0.07 | 131  | 34.8 | 0.02 | 0.06 | 1.99  | 0.02 | 4550  | 0.05 | 0.96 | 0.02 |
| 16B09_2 | 76   | 0.49 | 0.31 | 0.43  | 0.05 | 0.24 | 0.55  | 0.34  | 0.09 | 0.13 | 1.34  | 3.86 | 1.92 | bdl  | 131  | 31.3 | 0.03 | 0.11 | 1.99  | 0.02 | 3510  | 0.05 | 1.38 | 0.02 |
| 16B09_2 | 77   | 0.49 | 0.34 | 0.43  | 0.07 | 0.19 | 0.55  | 0.34  | 0.09 | 0.16 | 1.34  | 5.43 | 2.24 | 0.03 | 118  | 26.2 | 0.04 | 0.11 | 1.99  | bdl  | 2350  | bdl  | 0.24 | 0.03 |
| 16B09_2 | 78   | 0.49 | 0.25 | 0.43  | 0.05 | 0.10 | 0.55  | 0.34  | 0.05 | 0.11 | 1.34  | 2.62 | 1.90 | 0.09 | 98.0 | 31.0 | 0.02 | 0.07 | 1.99  | 0.01 | 4650  | 0.04 | 0.65 | bdl  |

| QCM            |            |  | Mn   | Co   | Ni*  | Ga   | Ge   | Se   | Sr** | Mo   | Ag*  | Cd    | In   | Sn   | Au** | Hg   | Tl   | Pb* | Bi   |
|----------------|------------|--|------|------|------|------|------|------|------|------|------|-------|------|------|------|------|------|-----|------|
| NIST SRM 610** | Mean       |  | 454  | 352  | 409  | 438  | 333  | 107  |      | 429  | 176  | 185   | 338  | 408  |      | 0.87 | 42.5 | 491 | 320  |
|                | SD         |  | 12.9 | 7.21 | 13.3 | 9.65 | 5.60 | 6.53 |      | 8.17 | 6.14 | 6.65  | 9.30 | 9.93 |      | 0.04 | 1.22 | 147 | 8.18 |
|                | Ref. Value |  | 444  | 410  | 458  | 433  | 447  | 138  |      | 417  | 251  | 270   | 434  | 430  |      | N/A  | 59.6 | 426 | 384  |
|                | 95% CL     |  | 13   | 10   | 4    | 13   | 78   | 42   |      | 21   | 9    | 16    | 19   | 29   |      | N/A  | 2.8  | 1   | 26   |
| MUL-ZnS-1*     | Mean       |  | 683  | 277  |      | 629  | 513  | 233  | 156  | 82.4 |      | 11272 | 516  | 1263 | 1.71 | 353  | 4.54 |     | 55.1 |
|                | 1 SD       |  | 23.2 | 10.4 |      | 66.9 | 58.9 | 9.00 | 3.88 | 7.09 |      | 278   | 11.6 | 53.8 | 0.80 | 315  | 0.13 |     | 1.27 |
|                | Ref. Value |  | 611  | 308  |      | 602  | 505  | 200  | N/A  | 59.8 |      | 11400 | 515  | 1048 | N/A  | N/A  | 11.1 |     | N/A  |
|                | 1 SD       |  | 42.7 | 7.70 |      | 13.8 | 32.3 | 2.00 | N/A  | 5.39 |      | 0.02  | 13.9 | 83.9 | N/A  | N/A  | 0.36 |     | N/A  |



|             |            |  |  |      |  |  |  |     |  |      |  |  |  |      |  |  |      |  |
|-------------|------------|--|--|------|--|--|--|-----|--|------|--|--|--|------|--|--|------|--|
| USGS MASS-1 | Mean       |  |  | 93.5 |  |  |  | bdl |  | 47.4 |  |  |  | 53.7 |  |  | 78.0 |  |
|             | 1 SD       |  |  | 2.77 |  |  |  | bdl |  | 0.74 |  |  |  | 7.54 |  |  | 2.83 |  |
|             | Ref. Value |  |  | N/A  |  |  |  | N/A |  | 67.0 |  |  |  | 47.0 |  |  | 75.5 |  |
|             |            |  |  | N/A  |  |  |  | N/A |  | 6    |  |  |  | 41   |  |  | N/A  |  |

For NIST SRM 610 the preferred values reported in the GeoReM database (<http://georem.mpch-mainz.gwdg.de>) (Jochum et al. 2005; 2011) were taken. The reference values for USGS MASS-1 were taken from Wilson et al. (2002) and the reference values for MUL-ZnS-1 were taken from Onuk et al. (2017).

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